Self-Study DA2

Evaluation of Programs for the Unemployed

Part I — Selection on Observables

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1. Data Cleaning

The raw dataset contains 12,000 observations and 18 variables. Several cleaning steps were required before it could be used for causal analysis. Inconsistent labels in `region` and `education` were harmonized, mapping multiple spellings to standard categories (A/B/C and low/medium/high/tertiary). Clearly invalid values were corrected: `takeup`$=$ 99 and `unemp_duration`$=$ 999 were treated as placeholders and recoded as missing; ages outside 18–67 and language scores outside 0–100 were also set to missing. Binary indicators with values outside $\{0, 1\}$ were likewise recoded as missing.

Three post-treatment variables were excluded from the covariate set: `active_participation`, `course_hours_completed`, and `diploma_awarded`. Conditioning on them would introduce post-treatment bias, since they are determined after treatment take-up and would absorb part of the treatment effect.

Additionally, 27 individuals in Region A who participated without receiving an offer (`offer`$=0$, `takeup`$=1$) were dropped. These cases are difficult to reconcile with the intended offer-based design and could weaken the interpretation of the IV strategy by blurring the distinction between compliers and always-takers. A complete-case sample was then formed by dropping observations with missing values on treatment, outcome, or the 10 pre-treatment covariates, leaving 9,608 observations out of the original 12,000. This reduction is driven mainly by the variables that were recoded as missing. Extreme but plausible values in income and outcomes were retained, while only clearly impossible values were removed.

2. Exploratory Data Analysis

Region A stands out institutionally. Its offer rate is 48.4%, whereas it is zero in Regions B and C, and take-up in Region A is largely driven by this offer mechanism. This indicates that treatment assignment in Region A follows a different process from the caseworker-based allocation observed in the other regions. Figure 1 illustrates this contrast.

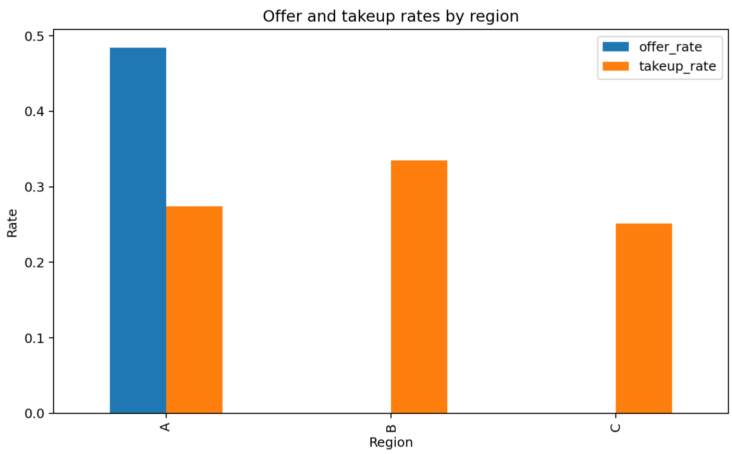

Figure 1: Offer and take-up rates by region.

Region C is the most problematic region for a pooled selection-on-observables analysis because it differs markedly from Regions A and B in observed characteristics. Table 1 shows that individuals in Region C are older on average, have much higher baseline income, shorter unemployment spells, and higher later outcomes. Figure 2 also shows a larger share of tertiary-educated individuals in Region C. These patterns suggest that Region C is structurally distinct in observable composition, making comparisons with the other regions less credible in a pooled analysis.

Table 1: Region summary statistics

	Region A	**Region B**	**Region C**
N	4,691	4,248	3,034
Offer rate	0.487	0.000	0.000
Take-up rate	0.269	0.335	0.251
Mean age	35.1	33.1	45.1
Female share	0.517	0.504	0.518
Mean baseline income	30,738	29,166	43,604
Mean unemp. months	5.1	8.1	4.1
Mean language score	63.7	62.9	65.9
Mean outcome	36,668	34,899	46,567

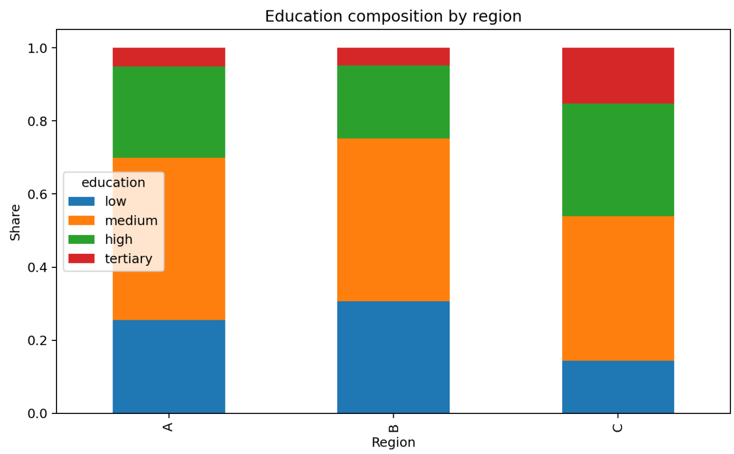

Figure 2: Education composition by region (stacked bar, share within region).

The propensity score overlap plots in Figure 3 show good common support in all three regions. Table 2 confirms that more than 99% of observations lie on common support in every region and that almost no treated units fall outside the support of the controls. Regions A and B therefore appear well suited for a selection-on-observables exercise from the point of view of overlap. Region C, however, remains less attractive because it is much less comparable to the other two regions in terms of covariate composition, even though overlap itself is good.

Table 2: Propensity score overlap statistics by region

	Region A	**Region B**	**Region C**
PS range (control)	[0.20, 0.60]	[0.05, 0.62]	[0.15, 0.40]
PS range (treated)	[0.20, 0.60]	[0.05, 0.62]	[0.15, 0.40]
% sample on common support	99.8%	100.0%	99.5%
% treated outside control	0.2%	0.0%	0.5%

Figure 4 complements the overlap analysis by reporting the absolute standardized mean differences

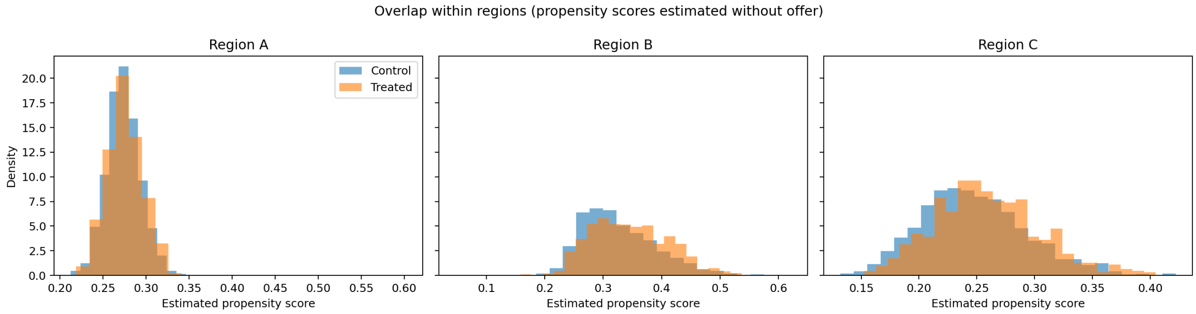

Figure 3: Propensity score overlap within each region (estimated without offer).

for the main pre-treatment covariates within each region. Region A still shows the best balance overall. Region B displays the largest residual imbalances, especially for unemployment duration, language score, and some education categories, with low education standing out most clearly in the balance plot. Region C also shows some imbalance, but the main concern there is not only within-region treated–control balance. Rather, as shown by the earlier descriptive evidence, Region C remains structurally different from Regions A and B in observable composition. Taken together, the results suggest two distinct issues: Region A is institutionally special because of the offer mechanism, whereas Region C is the least comparable region in terms of observable composition. For this reason, the main selection-on-observables analysis in Question 3 focuses on the pooled sample of Regions A and B.

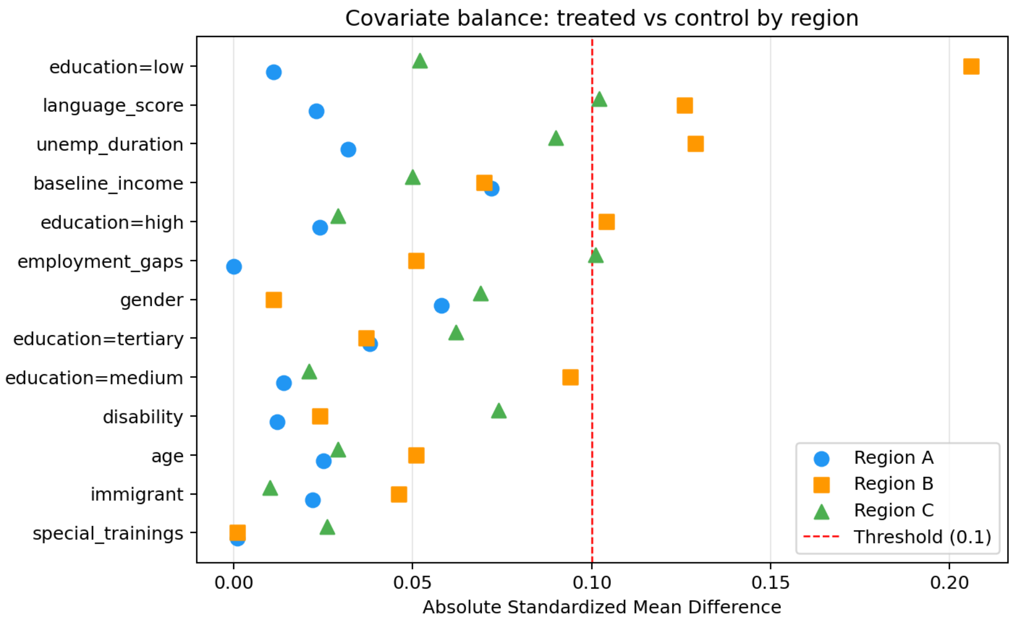

Figure 4: Absolute standardized mean differences (SMD) by covariate and region. Dashed line marks the 0.1 balance threshold.

3. Estimation under Selection on Observables

Although propensity score overlap is good in all three regions, Region C is excluded from the main estimation because it is much less comparable to Regions A and B in observed characteristics. The main selection-on-observables analysis therefore uses the pooled sample of Regions A and B. Since these two regions still differ in assignment mechanism, region is included as a covariate in all adjusted estimators to control for average observed differences between them. Treatment effects are estimated using four estimators; results are reported in Table 3.

Table 3: Treatment effect estimates in the pooled sample of Regions A and B

Method	Estimate	SE	z-stat
Raw mean difference	2208.711	168.953	13.073
Regression adjustment	2647.752	145.326	18.219
IPW	2731.677	991.316	2.756
IPW (normalized, bootstrap SE)	2674.375	145.021	18.441
Doubly robust	2644.605	143.273	18.458

All four estimates are positive and statistically significant at the 1% level. The raw mean difference is 2,208.7, while the adjusted estimates range from 2,644.6 to 2,731.7. Thus, once observable differences between participants and non-participants are taken into account, the estimated treatment effect becomes larger. This pattern is consistent with positive selection into treatment: the unadjusted comparison understates the effect relative to estimators that control for pre-treatment covariates. The adjusted estimates also cluster fairly closely, especially regression adjustment and doubly robust estimation.

The raw mean difference

$$\hat{\tau}^{raw} = \frac{1}{n_1}\sum_{i:D_i=1} Y_i - \frac{1}{n_0}\sum_{i:D_i=0} Y_i \tag{1}$$

has a causal interpretation only under unconditional exogeneity of treatment and SUTVA. That is not plausible here, because participation is voluntary and treatment assignment in the pooled A+B sample is not purely random. The raw estimate should therefore be interpreted mainly as a descriptive benchmark rather than as a credible causal effect.

Regression adjustment relies on the Conditional Independence Assumption (CIA), overlap, SUTVA, and correct specification of the outcome model:

$$\hat{\tau}^{RA} = \frac{1}{n}\sum_{i=1}^{n} \left(\hat{m}_1(X_i) - \hat{m}_0(X_i)\right) \tag{2}$$

Under CIA, treatment assignment is independent of potential outcomes conditional on the observed pre-treatment covariates. In this application, this means that after conditioning on the available covariates, there must be no remaining unobserved factor that jointly affects participation and later income.

Inverse probability weighting (IPW) relies on the same CIA, overlap, and SUTVA assumptions, but instead requires correct specification of the propensity score model:

$$\hat{\tau}^{IPW} = \frac{1}{n}\sum_{i=1}^{n} \frac{D_i Y_i}{\hat{p}(X_i)} - \frac{1}{n}\sum_{i=1}^{n} \frac{(1-D_i)Y_i}{1-\hat{p}(X_i)} \tag{3}$$

Its point estimate is again positive and similar in magnitude to the other adjusted estimates, but its standard error is much larger (991.3). This indicates that the IPW estimator is sensitive to large weights, which can arise when some estimated propensity scores are close to zero or one. Even when overlap is broadly acceptable, a small number of highly weighted observations can dominate the variance.

Doubly robust estimation

$$\hat{\tau}^{DR} = \frac{1}{n}\sum_{i=1}^{n} \left[\hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{p}(X_i)} - \frac{(1-D_i)(Y_i - \hat{m}_0(X_i))}{1-\hat{p}(X_i)}\right] \tag{4}$$

combines outcome regression and propensity-score weighting. It requires SUTVA, conditional ignorability and overlap, and it is consistent if either the outcome model or the propensity score model is correctly specified. Its estimate, 2,644.6, is almost identical to the regression-adjustment estimate.

Overall, the results point to a positive and economically meaningful effect of training participation on later income in the pooled sample of Regions A and B. Under their respective identifying assumptions, regression adjustment, IPW, and doubly robust estimation all target the ATE and deliver similar point estimates. The causal interpretation, however, still depends on CIA and overlap holding in this setting.

4. Credibility of the CIA

The Conditional Independence Assumption (CIA) cannot be tested directly, since it concerns unobserved determinants of treatment and potential outcomes. Its plausibility can only be assessed indirectly using overlap, covariate balance, and the broader institutional context. In this application, the most informative evidence comes from the within-region balance diagnostics in Figure 4 and from the similarity of the adjusted estimators in Question 3.

Region A shows the best balance overall, which is consistent with the fact that participation is closely linked to the offer mechanism. However, this does not automatically validate CIA for take-up. Randomization applies to the offer, not to the individual's decision to participate, and take-up may still depend on unobserved factors such as motivation or expected returns to training. CIA for treatment participation in Region A is therefore relatively more plausible than in a purely discretionary setting, but it is not guaranteed by the design alone.

Region B shows somewhat larger imbalance, especially on unemployment duration, language score, and some education categories, suggesting that caseworkers may have targeted the program toward more disadvantaged individuals along several observed dimensions. Conditioning on these observed characteristics should account for part of this selection, and the remaining covariates are still reasonably well balanced overall. CIA is therefore reasonably defensible in Region B after conditioning, though it remains uncertain.

Overall, CIA is relatively more plausible in Region A and reasonably defensible in Region B after conditioning, but not fully convincing in either case. More broadly, the main threat to CIA in this setting is that treatment assignment may depend on information unavailable in the dataset, such as motivation, employability, or local labour-market conditions. If such unobserved factors affect both participation and future earnings, CIA fails even after conditioning on all observed covariates. The estimates in Question 3 should therefore be interpreted as causal only under the maintained assumption that no important unobserved confounders remain once the available pre-treatment characteristics are taken into account.

5. Alternative identification strategies

A credible alternative strategy is an IV approach in Region A, using the randomized training offer as an instrument for actual participation.

Relevance. Region A is the only region with a non-zero offer rate, and take-up is substantially higher among those who received the offer. After excluding the 27 anomalous cases with `offer = 0` and `takeup = 1`, participation among non-offered individuals is essentially zero in the cleaned sample. This implies a strong first stage from offer to actual participation.

Independence. If offers in Region A were randomly assigned, the instrument is independent of individual-level unobservables such as motivation or employability. This makes the independence assumption more plausible here than in the non-pilot regions.

Exclusion restriction. The offer must affect later earnings only through actual participation. A possible threat is that receiving the offer may itself change behaviour — for example by increasing job-search effort or signalling institutional support — even among those who do not participate.

Monotonicity. The offer should not reduce participation for any individual. Under this assumption, the IV estimand identifies the effect for compliers, i.e. those who participate if and only if they receive the offer.

With a binary instrument $Z_i \in \{0, 1\}$ (the offer), the effect is identified by the Wald estimator:

$$\hat{\tau}_{LATE} = \frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]} \tag{1}$$

The numerator is the reduced-form effect of the offer on later earnings, while the denominator is the first-stage effect of the offer on participation. In the cleaned Region A sample, this first stage is large and positive, since participation is concentrated almost entirely among those who received the offer. This estimand is narrower than the ATE in Question 3, since it applies only to individuals whose participation is induced by the offer. Nevertheless, it is likely to be more credible in Region A because it does not rely on CIA, which is the assumption most threatened by unobserved motivation and caseworker discretion in this dataset.

Appendix: Additional Figures

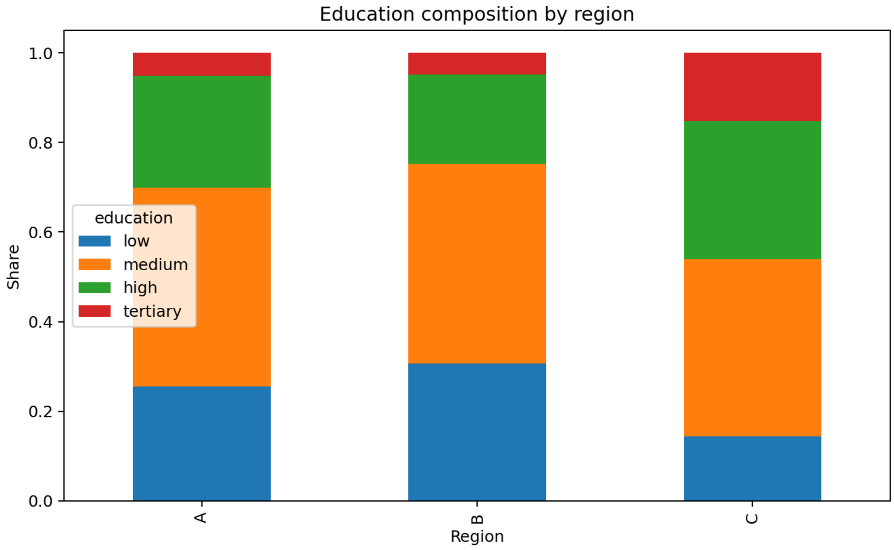

Figure 5: Education composition by region. Region C has a visibly larger share of tertiary-educated individuals, supporting the view that it is compositionally distinct from Regions A and B.

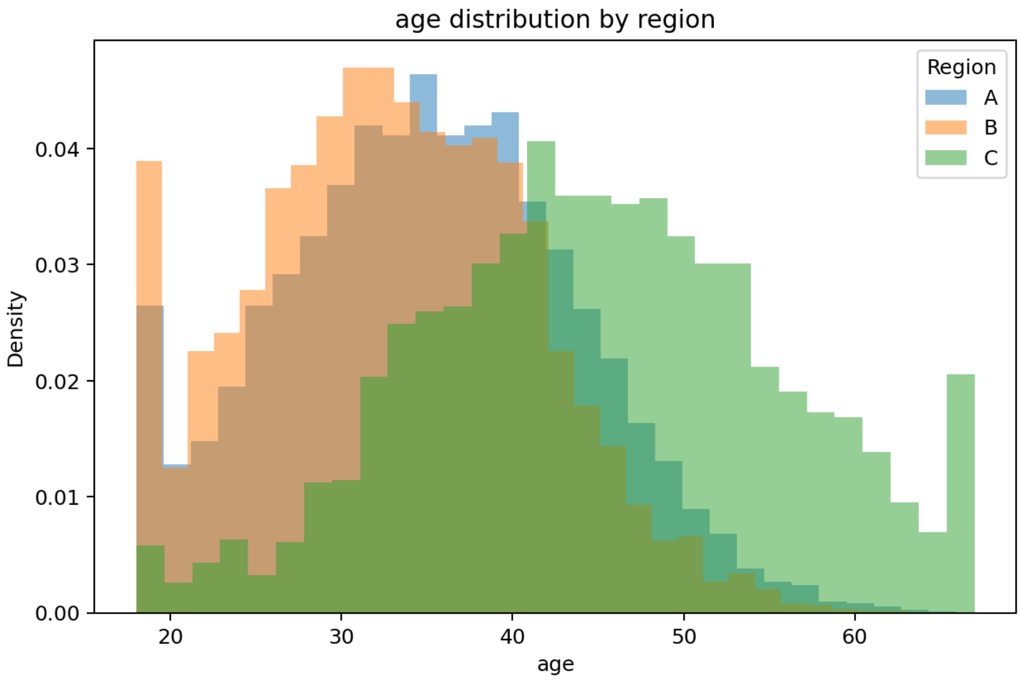

Figure 6: Age distribution by region. Region C is shifted to the right, indicating an older population on average than Regions A and B.

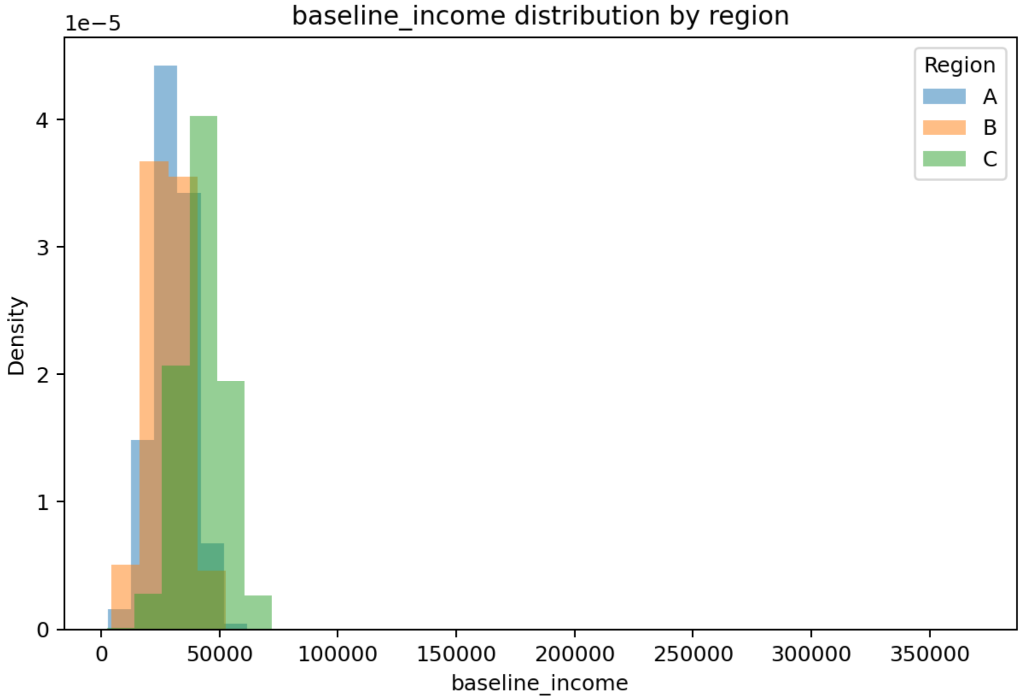

Figure 7: Baseline income distribution by region. Region C is concentrated at higher income levels, reinforcing its limited comparability with the other two regions in pooled analysis.

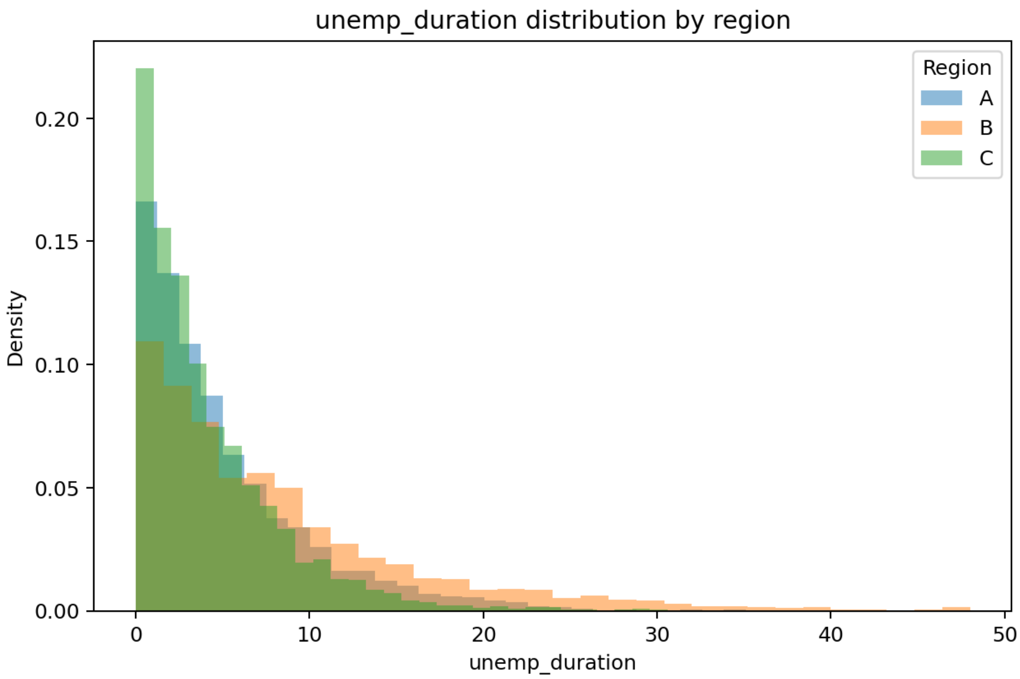

Figure 8: Unemployment duration by region. Region B shows a more dispersed distribution and longer unemployment spells, while Region C is more concentrated at shorter durations.